25901549 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCSTA204 - STATISTICAL MODELLING IN DATA SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **7** questions

Each question carries **2** marks

1. Define data. [K] / [CO1]
2. Mention two examples for transactional data. [U] / [CO1]
3. Mention one example each for discrete data and continuous data. [U] / [CO1]
4. Define nominal categorical data and ordinal categorical data. [U] / [CO1]
5. Mention one example for time series data. [U] / [CO1]
6. Define a test criterion. [K] / [CO2]
7. Define statistic and standard error [U] / [CO2]
8. What are the basic assumptions of one-way ANOVA? [An] / [CO3]
9. Define decision tree. [U] / [CO4]
10. The estimated regression equation is: [An] / [CO4]

$Y = 40 + 3X1 - 2X2$

where X1 = advertising cost and X2= price.

Estimate Y when X1 = 5 and X2 = 2.

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. Distinguish between semi structured data and unstructured data. [U] / [CO1]

12. Mention how data science is applied in targeting recommendation and delivery logistics. [U] / [CO1]

13. How spatial data is applied in environmental science and technology? [U] / [CO1]

14. Mahindra and Mahindra car company is claiming that their average sales for this quarter are 1500 cars, with SD 2. A sample study is taken from Karnataka and it was found that average sales from 90 showrooms was 1494. Is their claim valid? [A] / [CO2]

15. (1) What do you mean by goodness-of-fit testing? (2) Specify the chi-square statistic formula. (3) State the critical region of goodness-of-fit test. [U] / [CO3]

16. Mention the advantages and limitations of simple linear regression. [U] / [CO4]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. Explain biological data. [U] / [CO1]

18. A university administrator wants to know if a student's class rank is independent of whether they live on campus or off-campus. A random sample of 250 students provided the following data: Examine it using chi square test for independence. [A] / [CO3]

Living Status	Upper rank	Lower rank
On Campus	90	45
Off Campus	30	85

[12x1 = 12]